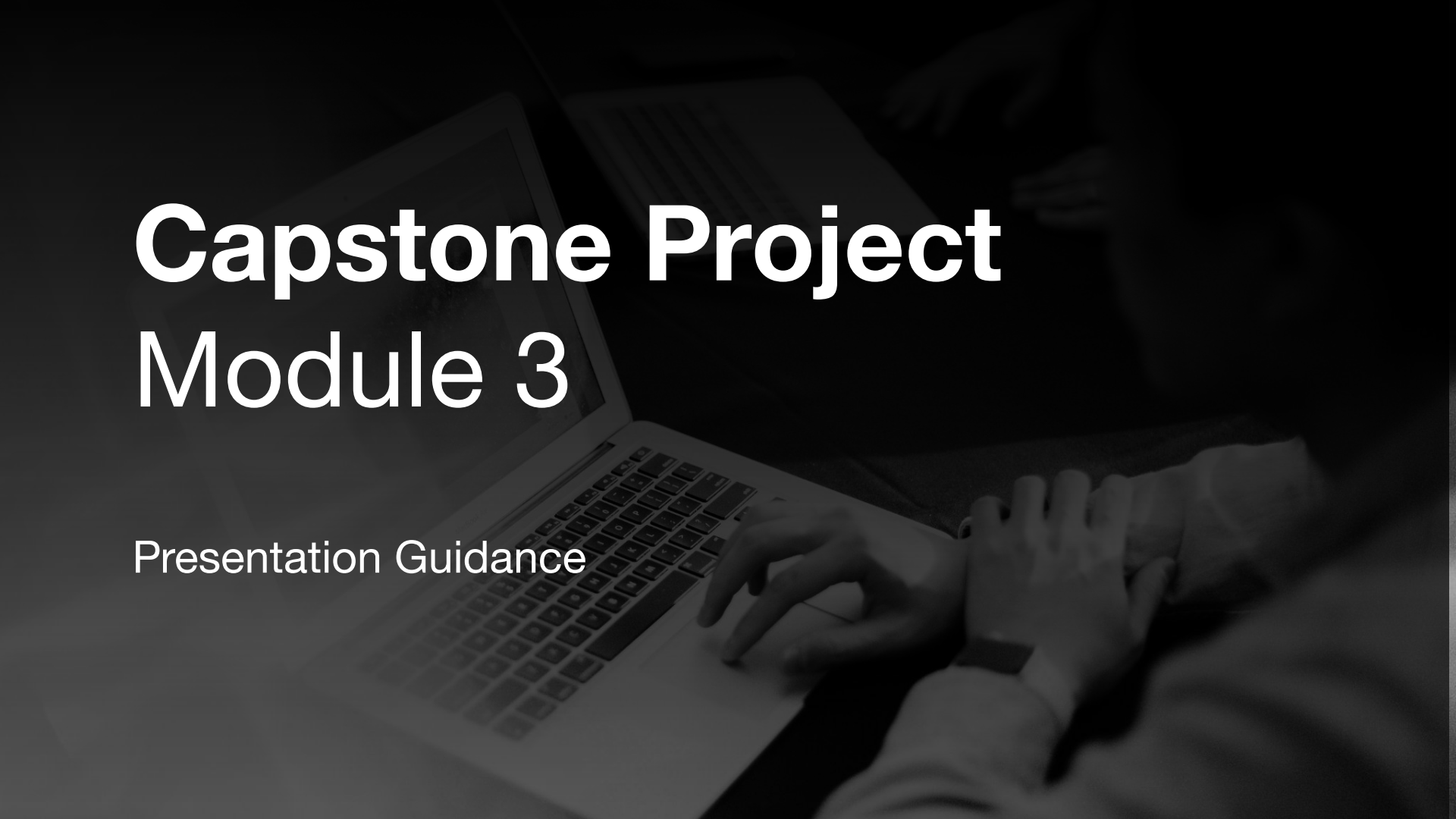


The background of the slide is a grayscale photograph of a person's hands typing on a laptop keyboard. The image is dimmed and has a dark, semi-transparent overlay applied to it, making the text stand out. The person's hands are positioned over the keyboard, and the laptop screen is visible in the upper left.

Capstone Project Module 3

Presentation Guidance

Notes

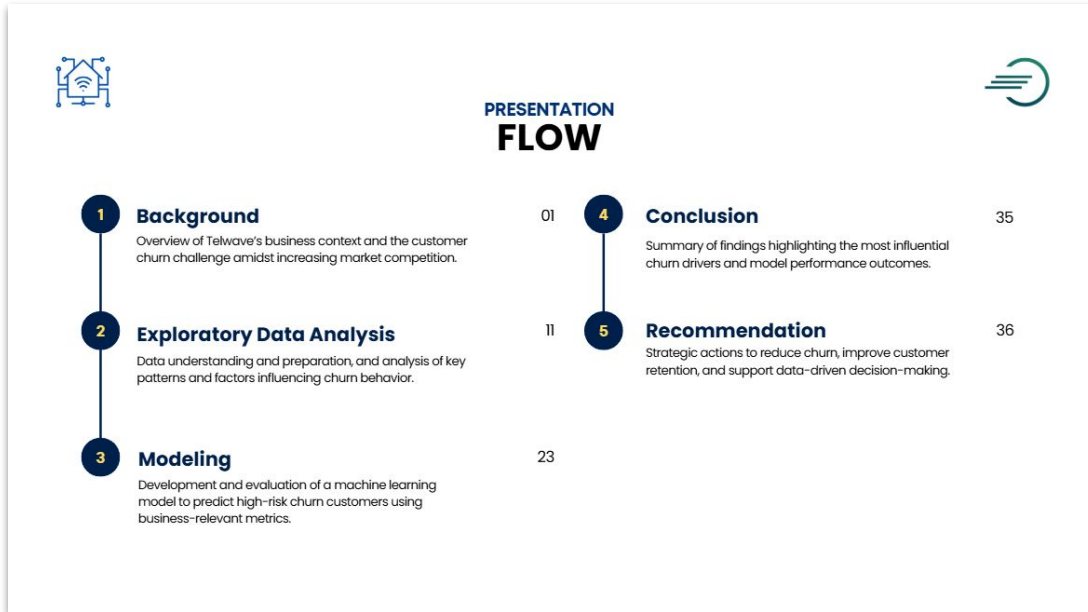
- Petunjuk Presentasi ini bersifat sebagai panduan, tidak untuk disalin mentah-mentah.
- Jumlah slide dibebaskan menyesuaikan delivery dari tiap peserta. Durasi ideal presentasi sudah dicantumkan pada Guideline Capstone Project Module 3.
- Perhatikan kembali Guideline Capstone Project Module 3 yang telah dibagikan sebelum mengerjakan project.

Cover



- Cover: to the point menjelaskan apa yang dianalisis
- Kalau ada gambar, ilustrasinya menjelaskan apa yang akan dianalisis

Outline



- Menjelaskan apa saja yang akan disampaikan

Executive Summary



Background

Telwave

Telwave is a **telecommunications company** that provides **internet, Wifi, and telephone services**. Telwave offers flexibility by allowing customers to choose services individually or as part of a bundled package that combines multiple services.

VISION
Connecting the world through superior, fast, and reliable telecommunications services.

7K+ Registered Customers **50+** County Covered



- **Background:** Tentang Company dan bagaimana bisnisnya berjalan (mencari revenue)
- **Problem:** Apa permasalahan yang dialami company sebelum menggunakan ML
- **Stakeholder/User:** Siapa yang membutuhkan hasil analisis dan prediksinya
- **Goal:** Definisi dari targetnya (misalnya Churn didefinisikan sebagai tidak login dalam 30 hari) dan kenapa harus pakai ML
- **Analytical Approach:** DA, ML (Classification/Regression)
- **Metric:** penjelasan metric yang dipilih itu apa dan alasannya kenapa pakai metric tsb
- Bisa diberi tambahan konteks statistik penting dari data yang akan dianalisis

Data Understanding



Exploratory Data Analysis - Dataset Overview 12

DATA UNDERSTANDING

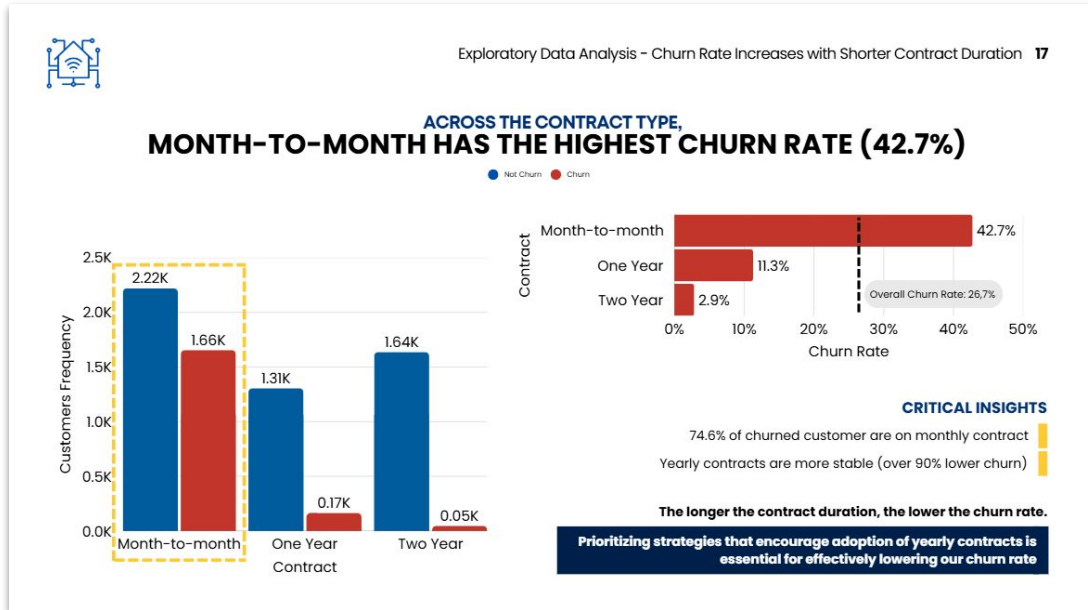
● Main Dataset ● Extra Dataset

*Each row in the dataset represents a unique customer, identified by the Customer ID column.

Customer Demographic		Service Characteristic	
CustomerID	Unique Identifier for each customer	PhoneService	Whether the customer has phone service (Yes/No)
Gender	The gender of the customer (Male/Female)	MultipleLines	Whether the customer has multiple phone lines (Yes/No/No phone service)
SeniorCitizen	Senior citizen status (Yes/No)	InternetService	Customer's internet service type (DSL, Fiber optic, No)
Age	Age of the customer in years	StreamingTV	TV streaming service (Yes/No/No internet service)
Married	Marriage status (1=Married, 0=No)	StreamingMovies	Movie streaming service (Yes/No/No internet service)
City	The city of the customer's residence	StreamingMusic	Music streaming service (1=Yes, 0=No)
Country	The country of the customer's residence	UnlimitedData	Unlimited data package (1=Yes 0=No)
ZipCode	The zip code of the customer's residence	Contract	Contract type (Month-to-month, 1 year, 2 year)
Population	The estimated population of the customer's zip code	PaperlessBilling	Whether the customer has paperless billing (Yes/No)
Latitude, Longitude	Geographical Coordinates	PaymentMethod	The customer's payment method
Partner	Whether the customer has a partner (Yes/No)		
Dependents	Whether the customer has dependents (Yes/No)		

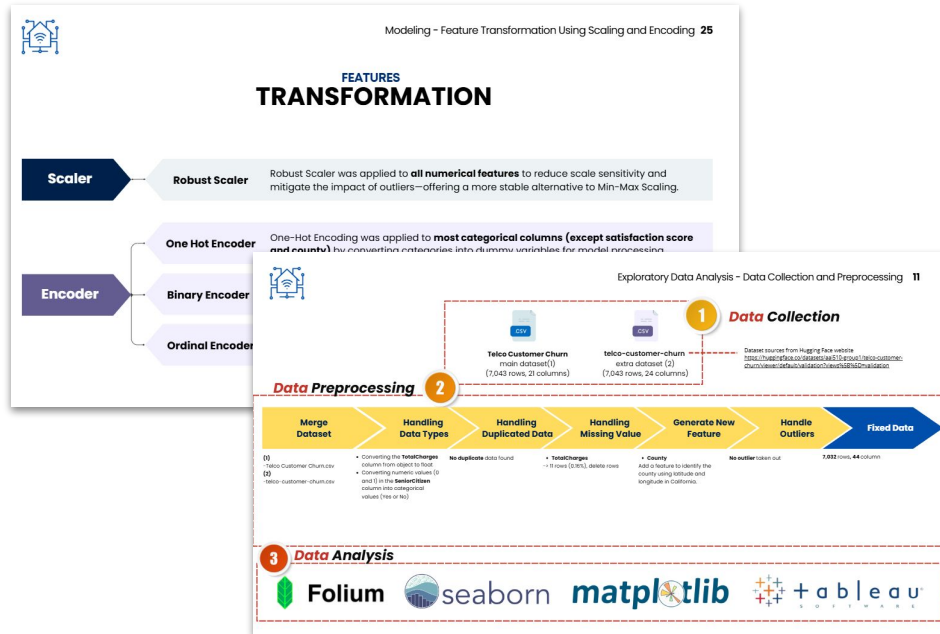
- Tiap observasi mewakili apa
- Tiap kolom artinya apa
- Apakah ada data external

Exploratory Data Analysis



- Distribusi data target
- Distribusi data features (yang perlu diperlihatkan saja)
- Bivariate/Multivariate: memperlihatkan hubungan feature vs target
- Hanya tampilkan yang menarik dan signifikan
- Grafik ideal dan clean, highlight bagian penting
- Tidak banyak grafik dalam 1 slide
- Insight utama langsung ada di judul
- Optional: bisa langsung kasih hint kemungkinan rekomendasi apa yang bisa dilakukan

Data Cleaning & Feature Engineering



- Tidak perlu tampilkan codingan
- Rangkum proses data cleaning
- Rangkum proses feature engineering
- Lebih baik tampilkan dalam bentuk flowchart
- Saat presentasi, sebutkan alasan kenapa menggunakan metode-metode yang dipilih

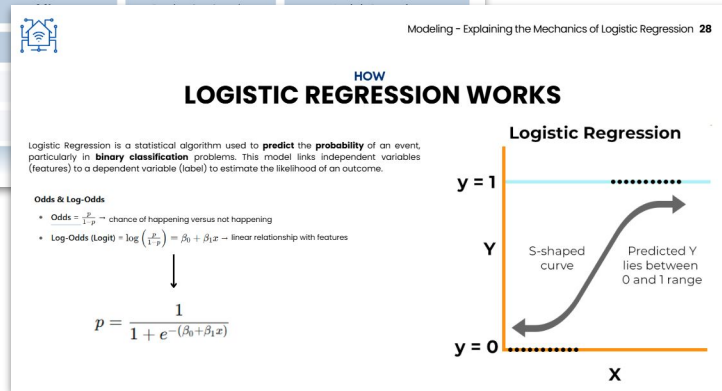
Modeling

Modeling - Base Model Results and Classifier Performance 27

BASE MODEL SUMMARY

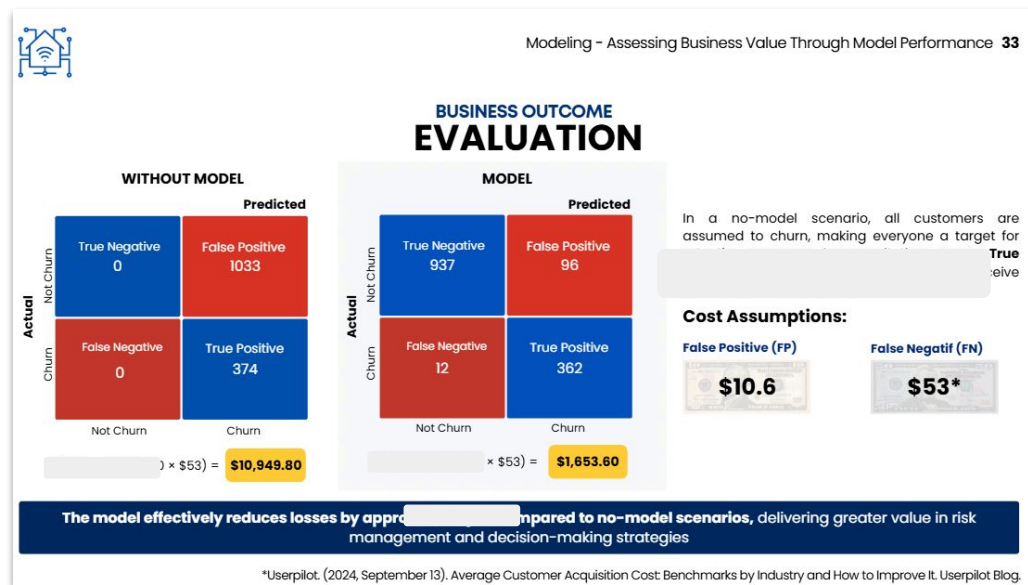
CV = 5

	F2 Mean Test Score	F2 Std Test Score	Resampling Method	Classification Method
1	0.931	0.01	RandomUnderSampler	Logistic Regression
2	0.930			
3	0.928			
4	0.925			
5	0.921			



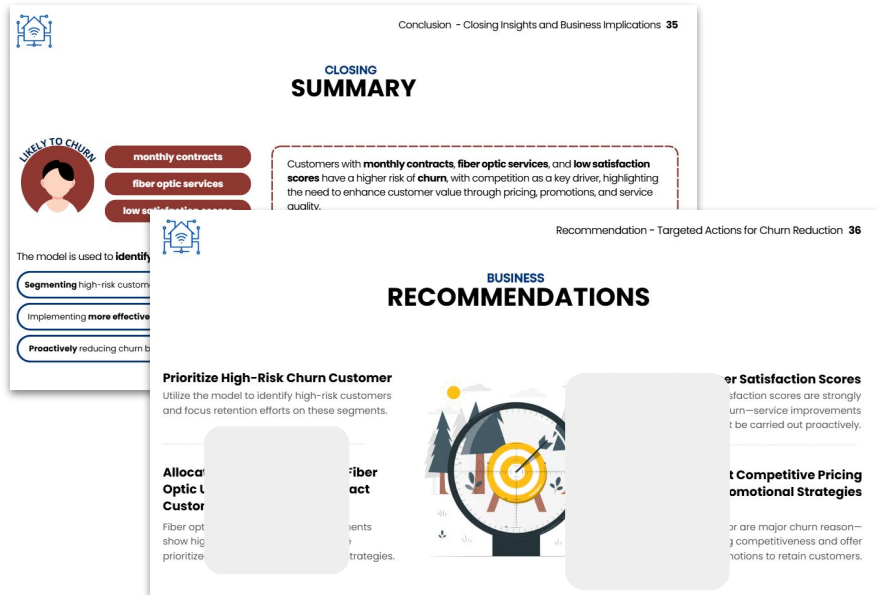
- Tampilkan hasil pemilihan model dari cross validation
- Jelaskan cara kerja algoritma dari final model
- Jelaskan hyperparameter dari final model
- Jelaskan interpretasi model atau explainable dari model (jika modelnya non-interpretable)
- Jelaskan limitasi model: akurat pada rentang data seperti apa, feature yang bisa ditambah apa, alternatif metode lain yang bisa menghasilkan output yg lebih akurat, dll.

Business Impact

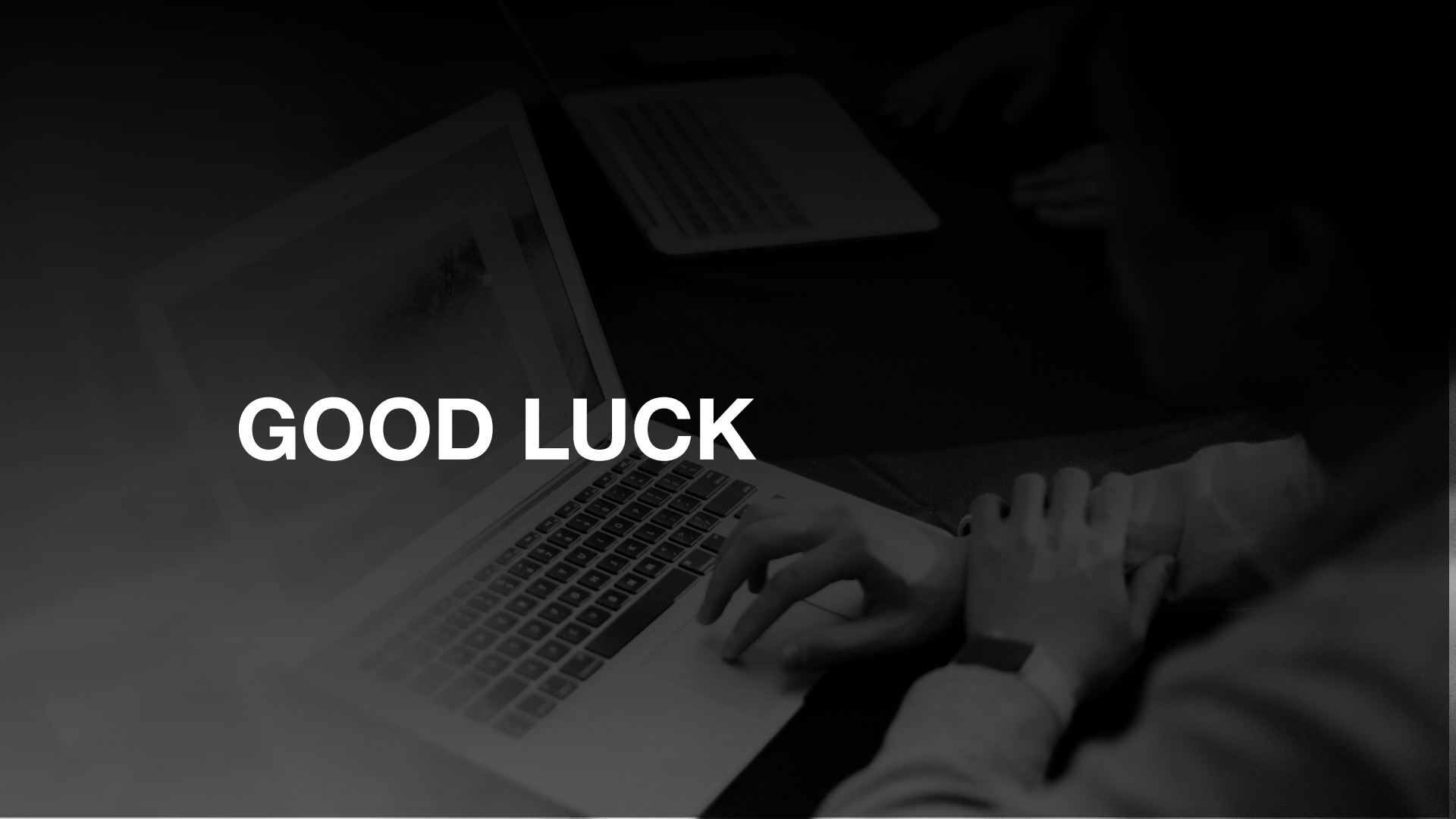


- Jelaskan keadaan (cost/revenue/time) sebelum vs sesudah menggunakan ML

Conclusion & Recommendation



- Conclusion merangkum hasil analisis baik diagnostic ataupun predictive
- Recommendation harus berdasarkan hasil analisis, tidak ngambang.
- Cari referensi rekomendasi bisnis di internet sesuai dengan domain/business yang dikerjakan



GOOD LUCK